Arjuna JEE 2027

Maths by Amarnath Anand Sir

Quadratic Equations

DPP: 7

Total Duration - 45 Min

Q1 The number of all possible positive integral values of α for which the roots of the quadratic equation, $6x^2 - 11x + \alpha = 0$ are rational numbers, is equal to

(A) 3
(B) 2
(C) 4
(D) 5

Q2 The least integral value of k for which the equation $x^2 - 2(k+2)x + 12 + k^2 = 0$ has two distinct real roots is

(A) 0
(B) 2
(C) 3
(D) 4

Q3 If $a + b + c = 0$, then the roots of the equation $4ax^2 + 3bx + 2c = 0$ are

(A) Equal
(B) Imaginary
(C) Real
(D) None of these

Q4 If the difference between the roots of the equation $x^2 + ax + 1 = 0$ is less than $\sqrt{5}$, then the set of possible values of a is

(A) $(-3, 3)$
(B) $(-3, \infty)$
(C) $(3, \infty)$
(D) $(-3, -\infty)$

Q5 If the roots of the equation $qx^2 + px + q = 0$ where p, q are real, be complex, then the roots of the equation $x^2 - 4qx + p^2 = 0$ are

(A) Real and unequal
(B) Real and equal
(C) Imaginary
(D) None of these

Q6 If one root of the equation $x^2 + px + 12 = 0$ is 4, while the equation $x^2 + px + q = 0$ has equal roots, then the value of 'q' is:

(A) 4
(B) 12
(C) 3
(D) $\frac{49}{4}$

Q7 If the equation $\left(k^2 - 3k + 2\right) x^2 + \left(k^2 - 5k + 4\right) x + \left(k^2 - 6k + 5\right) = 0$ is an identity then the value of k is

(A) 1
(B) 2
(C) 3
(D) 4

Q8 The roots of the equation $(b - c)x^2 + (c - a)x + (a - b) = 0$ are

(A) $\frac{c-a}{b-c}, 1$
(B) $\frac{a-b}{b-c}, 1$
(C) $\frac{b-c}{a-b}, 1$
(D) $\frac{c-a}{a-b}, 1$

Q9 If α, β are the roots of the equation $ax^2 + bx + c = 0$, then $\frac{\alpha}{a\beta+b} + \frac{\beta}{a\alpha+b} =$

(A) $2/a$
(B) $2/b$
(C) $2/c$
(D) $-2/a$

Q10 If the equation $px^2 + 2qx + r = 0$ and $px^2 + 2rx + q = 0 (q \neq r)$ have a common root, then $p + 4q + 4r$ equals

(A) 0
(B) 1
(C) 2
(D) -2

Q11 The equations $ax^2 + bx + c = 0$ and $bx^2 + cx + a = 0$, where $b^2 - 4ac \neq 0$ have a common root, then $a^3 + b^3 + c^3$ is equal to $(a \neq 0)$

(A) $3abc$
(B) abc
(C) 0

(D) 1

Q12 If $x^2 + ax + 10 = 0$ and $x^2 + bx - 10 = 0$, have a common root, then $a^2 - b^2$ is equal to

(A) 10 (B) 20

(C) 30 (D) 40

Q13 If the quadratic equation $3x^2 + ax + 1 = 0$ and $2x^2 + bx + 1 = 0$, have a common real root, then the value of the expression $5ab - 2a^2 - 3b^2$ is

(A) 0 (B) 1

(C) -1 (D) none of these

Q14 If $x^2 - 2x + c = 0$ and $x^2 - ax + b = 0$ have a root in common and the second equation has equal roots, then $b + c$ is equal to

(A) a (B) $2a$

(C) $3a$ (D) None of these

Q15 If the equation $x^2 - (p+q)x + 4 = 0; x^2 - 2qx + 8 = 0$ and $x^2 - (2p+q)x + 12 = 0$ have exactly one common root then which of the following is(are) CORRECT?

(A) $p = 2$

(B) $q = 3$

(C) Common root is 4

(D) Sum of all uncommon root is 6

Q16 If the equation $ax^2 + bx + c = 0$ has distinct real roots, both negative, then-

(A) a, b, c must be of same sign

(B) a, b must be of opposite sign

(C) a, c must be of opposite sign

(D) a, b must be of same sign and opposite to sign of C

Q17 If $x^2 + 2ax + 10 - 3a > 0$ for all $x \in R$, then

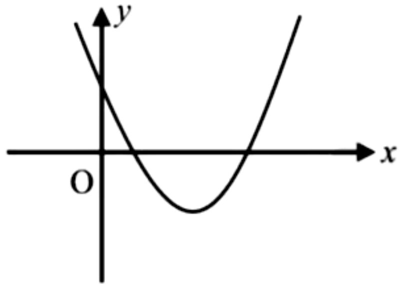
(A) $-5 < a < 2$

(B) $a < -5$

(C) $a > 5$

(D) $2 < a < 5$

Q18 The graph of $y = ax^2 + bx + c$ is shown in figure. Which of the following does **NOT** hold good?



(A) $ab^2c^3 > 0$

(B) $ab^3c^2 < 0$

(C) $ab^3c^5 > 0$

(D) $b^2 > 4ac$

Q19 The integer $'k'$, for which the inequality $x^2 - 2(3k-1)x + 8k^2 - 7 > 0$ is valid for every x in R, is

(A) 4 (B) 2

(C) 3 (D) 0

Q20 The equation $\frac{a(x-b)(x-c)}{(a-b)(a-c)} + \frac{b(x-c)(x-a)}{(b-c)(b-a)} + \frac{c(x-a)(x-b)}{(c-a)(c-b)} = x$ is satisfied by

(A) No value of x

(B) Exactly two values of x

(C) Exactly three values of x

(D) All values of x

Answer Key

Q1	(A)
Q2	(C)
Q3	(C)
Q4	(A)
Q5	(A)
Q6	(D)
Q7	(A)
Q8	(B)
Q9	(D)
Q10	(A)
Q11	(A)
Q12	(D)
Q13	(B)
Q14	(A)
Q15	(A, B, C, D)
Q16	(A)
Q17	(A)
Q18	(C)
Q19	(C)
Q20	(D)